2.6.2. Concept of a Mining Cycle Vis-à-vis Web Cutting by a Shearer

The support reaction of the shield and the corresponding leg closure are the results of the interaction between roof, coal seam floor and shield support. Based on the pattern of pressure variation shown above and in the earlier study, four distinctive periods define a mining cycle. The four periods of a mining cycle, shield pressure variation versus time, are shown in Figure 2.9 (Deb *et al.*, 2006).

A mining cycle begins after setting the shield and is designated by Ps. After the pressure rapidly increases between a and b, termed the rapid increase period, a long and relatively stable support resistance is achieved between b and c. This happens since the shearer cuts coal away from that shield. This period is termed a relatively stable period, tb. Support resistance increases rapidly, giving the cutting influenced period, tc between c and d. Finally, as the neighbouring supports are disengaged and advanced, the roof load suddenly transferred to the adjacent shield causing neighbouring shield movement period, td, between d and e. At this time, the final or the maximum pressure is achieved. Here, Pf is the final pressure, and ΔP is the developing pressure from setting to final pressures. These mining cycle signatures vary with the type of roof, such as weak/fractured roofs to stable or strong roofs (Deb *et al.*, 2006).

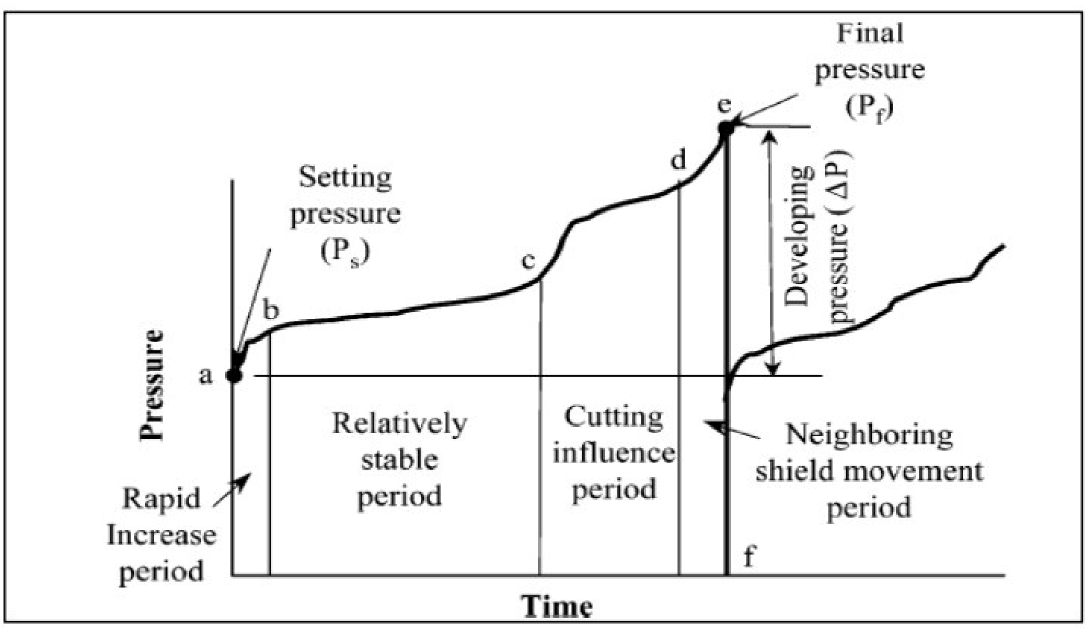

Figure 2.9. Mining cycle of a longwall shield Support (Deb *et al.*, 2006)